

# 3. Digital Signal Processing

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# 3.1. Sound representation



Analogue  
Bit depth  
Bit rate  
Sample rate  
Compression

## Discussion

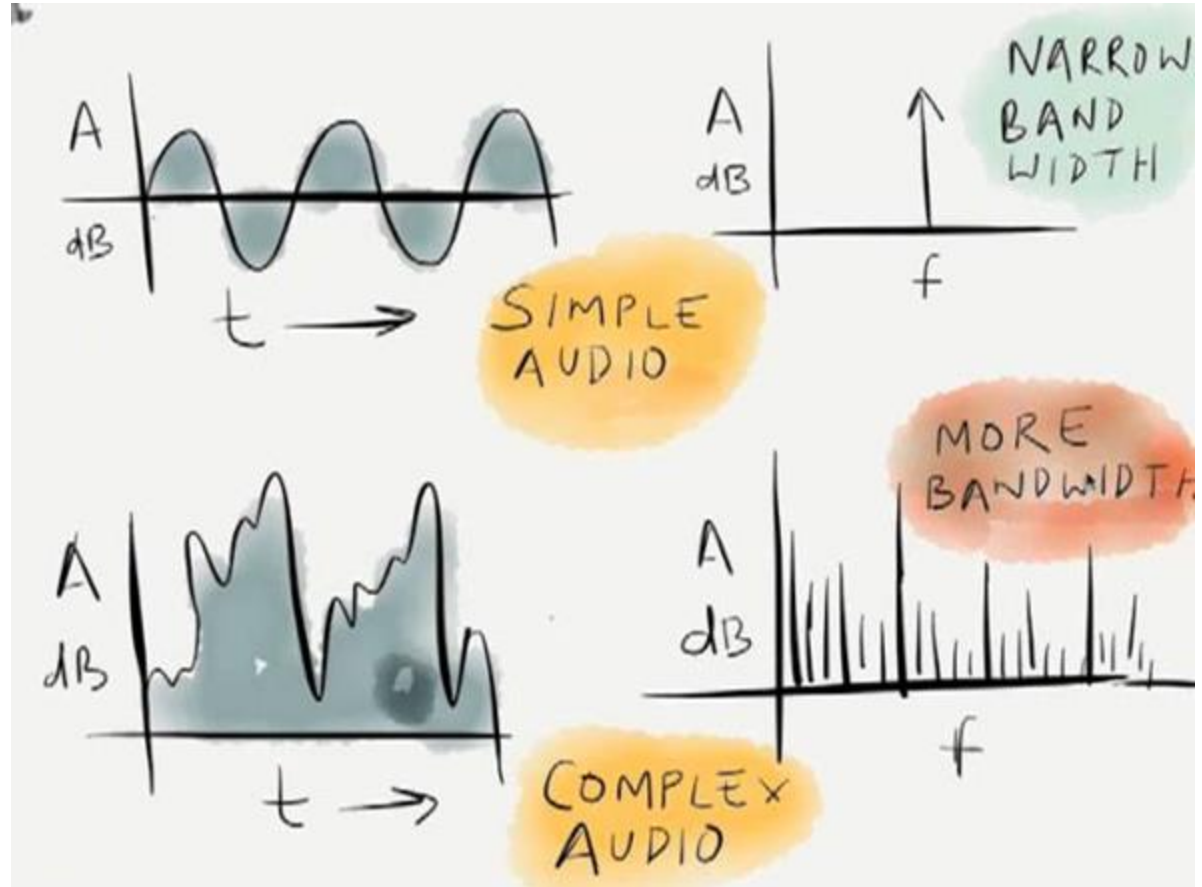
Which of the following has the best quality?

**Record, CD or MP3?**

# Analogue Digital Converter (ADC)

- Sampling
- Quantization

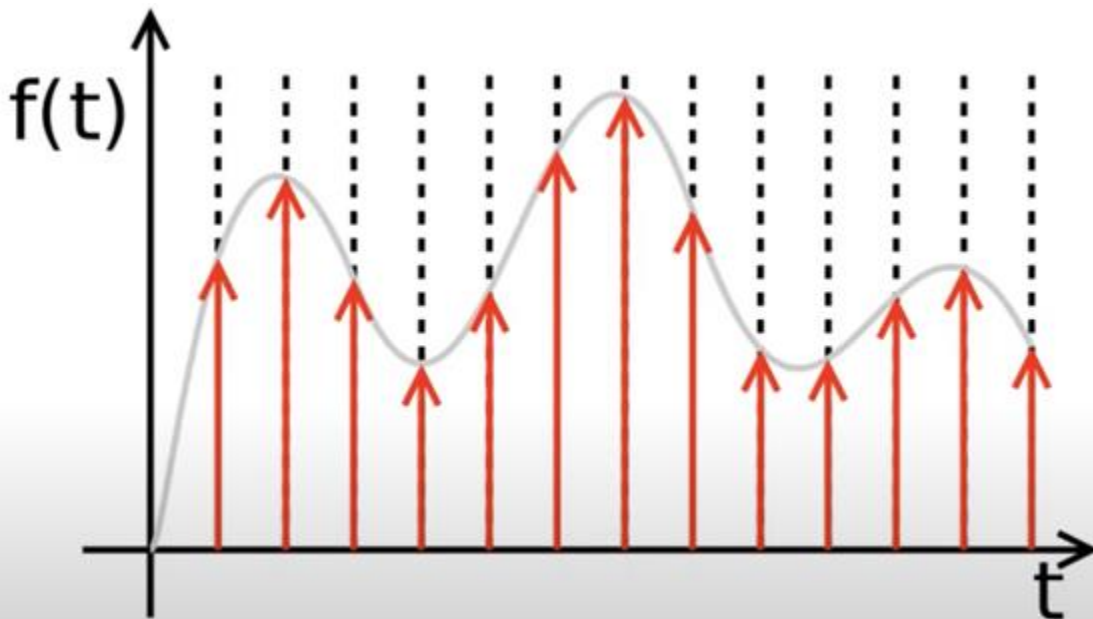
**Analogue:** Continuous changing values



### 3.1.1. Sampling

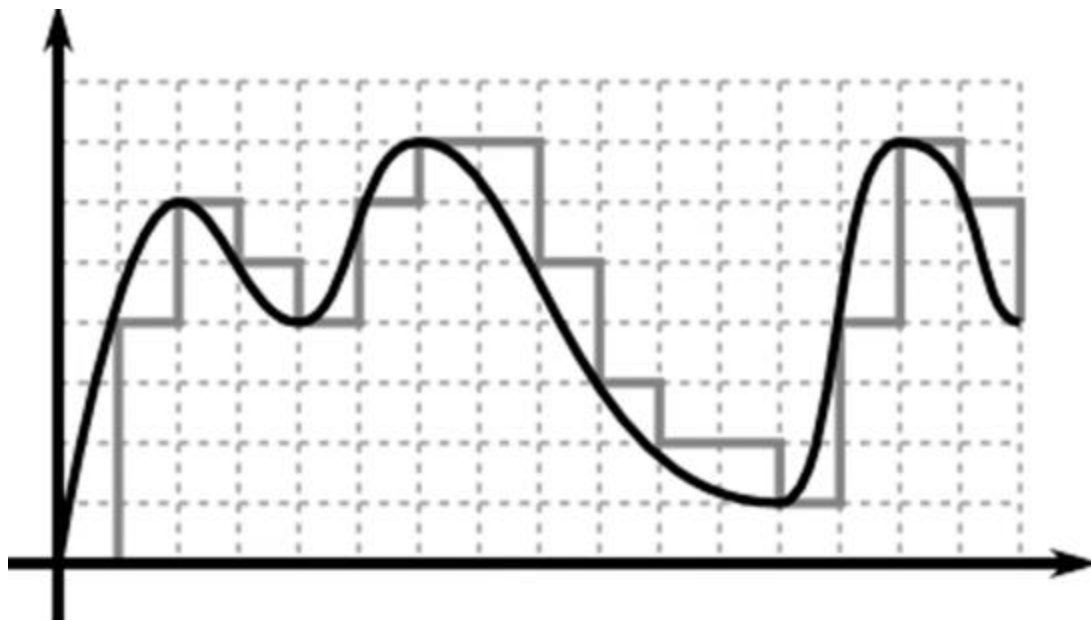
- **Sampling:** Converting a continuous analogue time signal into a discrete time representation
  - Carried out at key points → sampling intervals

A small **sampling interval** results which results in a better **quality sound** file and increases the **size** of the **audio** file



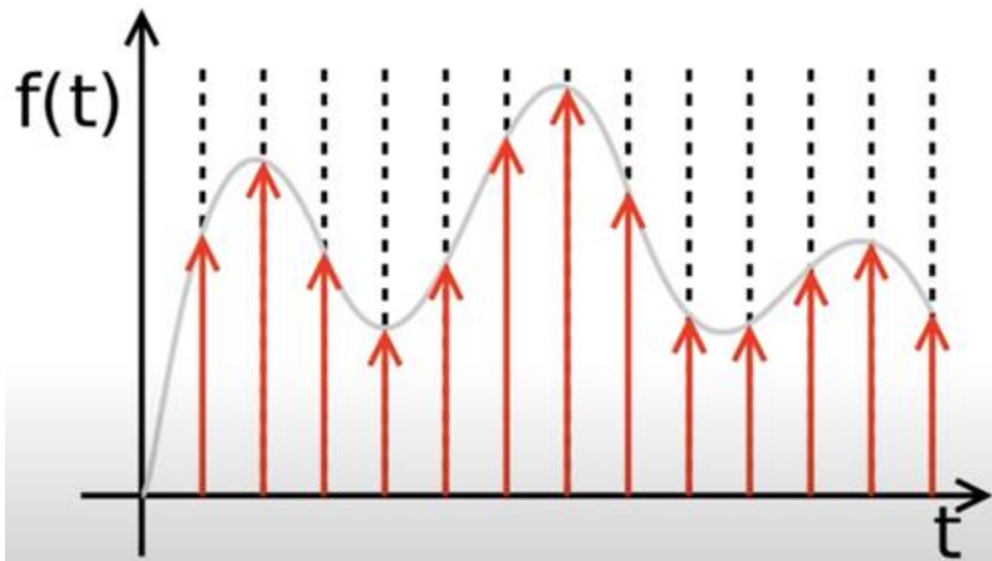
## Approximate in Sampling

The sound quality will be poorer and less smooth because some bits are missing.

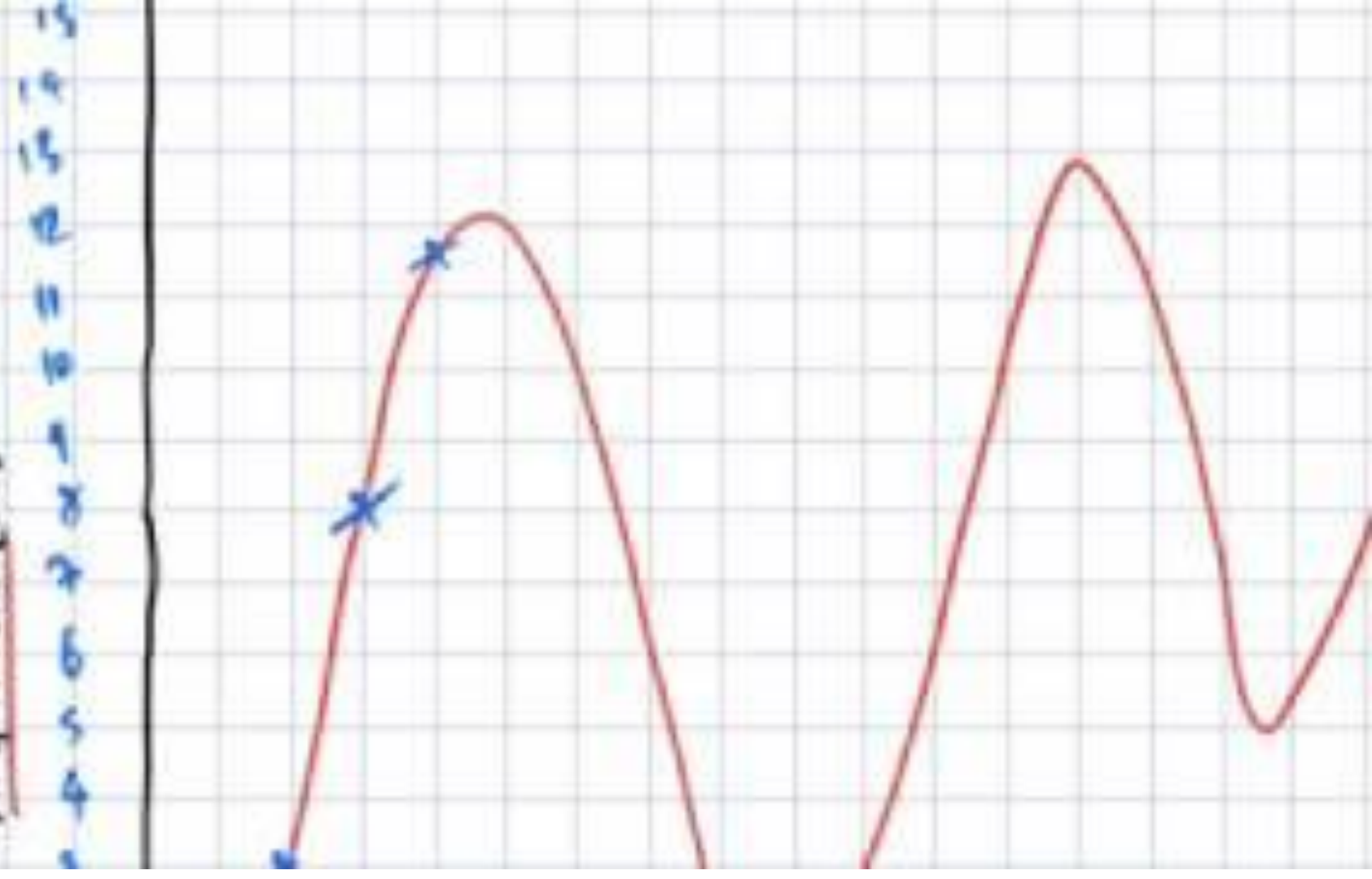


## Sample rate

- Number of times analogue sound is sampled per second to transform to discrete digital signal
- Higher ~ more samples ~ Higher quality ~ bigger file size
  - Small sampling interval
- Common unit: **kHz**  
(thousands of samples per second)
- CD: 44.1 kHz

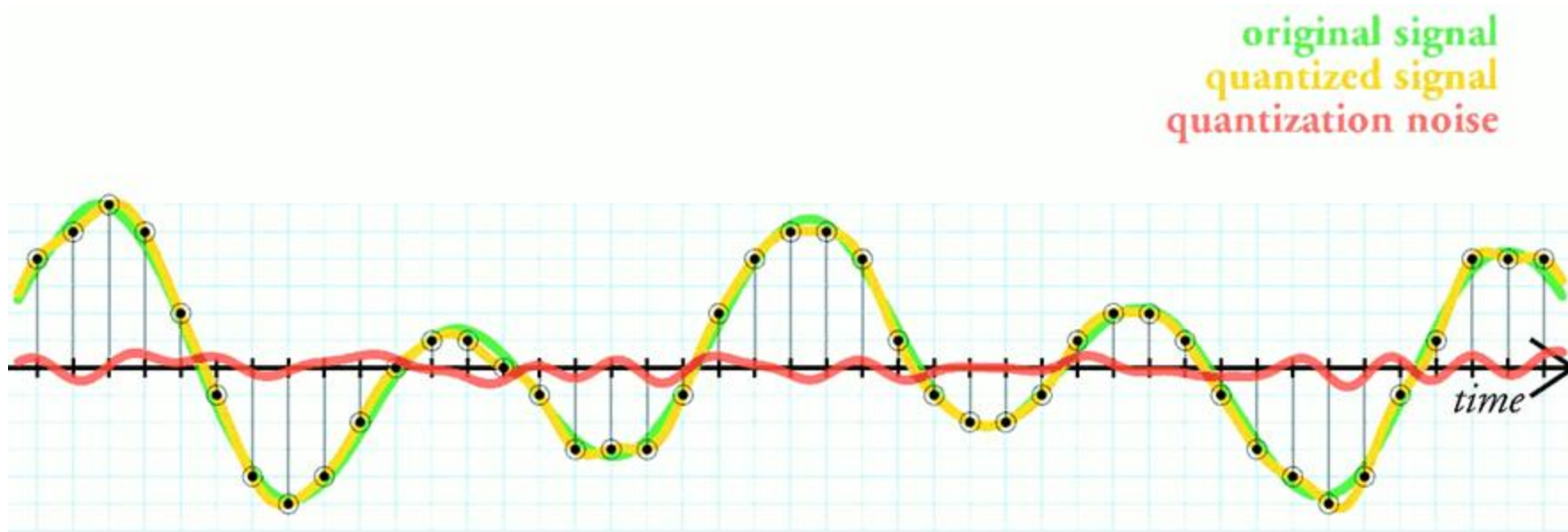


Amplitude (dB)



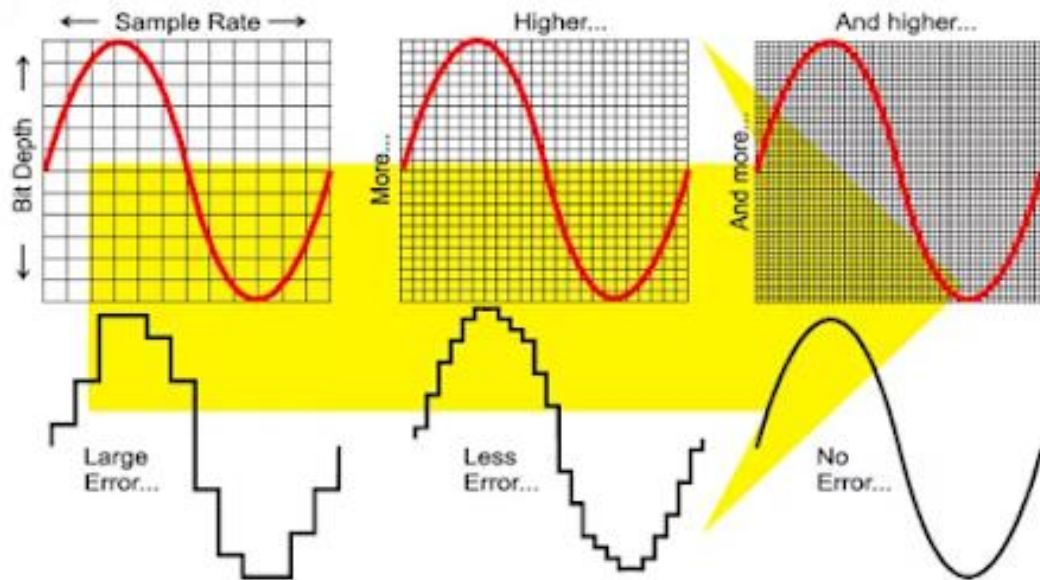
### 3.1.2. Quantization

- Mapping the **continuous amplitude values** of an **analog** audio signal into a **discrete** set of values in digital format, after sampling



# Bit Depth vs Bit Rate

- **Bit depth** is the **number** of **bits** that are used to **represent each sample** of audio
- **Bit rate** is the **number** of **bits** used to **represent one second** of audio (bits per second)
  - **Bit rate = Bit depth**  
**x Sample rate**
  - A higher bit rate means better audio quality and, thus, a larger file size.



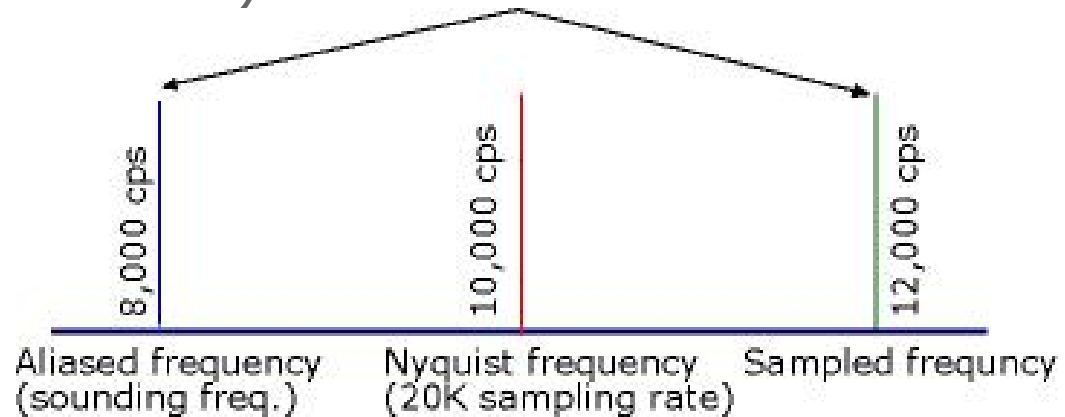
# Nyquist Theorem

## or **Nyquist-Shannon Sampling Theorem**

- fundamental principle in signal processing that explains how to **accurately** convert a continuous analog signal into a digital signal
- “to faithfully reconstruct a signal from its samples, the sampling rate must be at least twice the highest frequency present in the signal”.
  - E.g. Frequencies up to 20 kHz (highest), the sampling rate must be at least 40 kHz.

# Nyquist Rate

- Maximum frequency that can be accurately represented, which is half the sampling rate
  - If not → Aliasing → Distortion
  - E.g. Audio CDs use a sampling rate of 44.1 kHz to capture frequencies up to 22.05 kHz, covering the full range of human hearing (20 Hz–20 kHz).



## Hertz

1Hz = 1 Sample Per Second

1000Hz = 1000 Samples Per Second

44100Hz = 44,100 Samples Per Second



# File Size and Sample Rate, Bit Depth

## Calculating Sound File Size

Bits Per Second = Sample Rate x Bit Depth

File Size in Bits = Bits Per Second x Duration

File Size in Bytes = Number of Bits ÷ 8

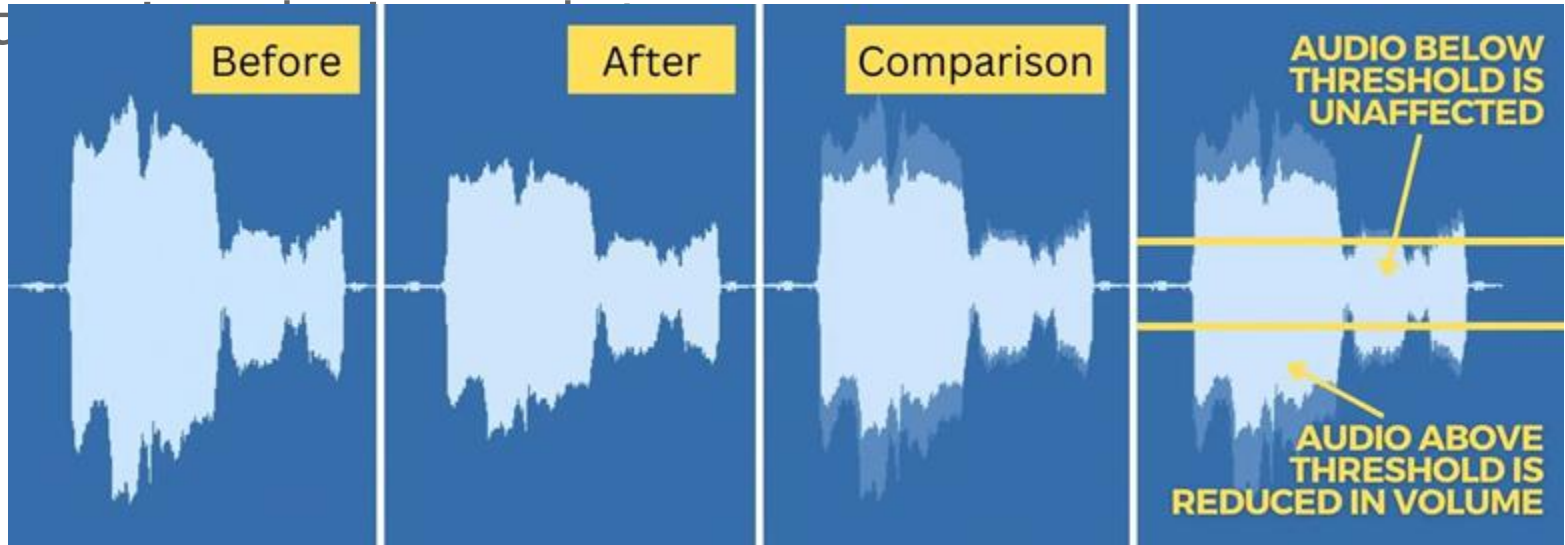
File Size in KB = Number of Bytes ÷ 1024

### 3.1.3. Compression

| Feature       | Lossy Compression            | Lossless Compression        | Uncompressed                           |
|---------------|------------------------------|-----------------------------|----------------------------------------|
| File Size     | 70-90% small                 | 30-70% smaller              | Very large                             |
| Audio Quality | Reduced (depends on bitrate) | Original Quality            | Original Quality                       |
| Use Case      | Streaming, mobile storage    | Archiving, professional use | Audiophile listening, professional use |
| E.g.          | MP3, AAC, OGG, Opus          | FLAC, ALAC                  | WAV, AIFF                              |

# Lossy Compression of Audio

- Permanently **removing** certain audio **information** that is deemed **less important** or **less noticeable** to the human ear.
- Balances file size and sound quality, making it ideal for streaming



# Interest Facts on Bit Rate

## 1. Music Streaming

- 96 kbps: Low quality, often used for streaming on mobile devices with limited data
- 128 kbps: Standard quality, commonly used in MP3 files for general music playback
- 256 kbps: High quality, used in services like Apple Music or high-quality MP3 files
- 320 kbps: Very high quality, near-CD quality, preferred by audiophiles for MP3 playback

## 2. Lossless Audio (Uncompressed Formats)

- 1,411 kbps (CD quality): Audio from CDs uses a bitrate of 1,411 kbps (44.1 kHz sample rate, 16-bit depth, stereo)
- 4,608 kbps: High-resolution lossless audio, typically 96 kHz sample rate, 24-bit

# Interest Facts on Bit Rate

## 3. Video Streaming Audio

- 96-160 kbps: Used for audio tracks in standard-definition (SD) video streaming
  - e.g. 128kbps for YouTube SD, 96 kbps for basic stereo audio track on Netflix
- 192-256 kbps: Common for high-definition (HD) streaming audio tracks
  - e.g. 192kbps for YouTube HD
- 384 kbps or higher: Used in ultra-high-definition (4K) video streaming for enhanced audio quality
  - e.g. 768 kbps or higher for Dolby Atmos surround sound.

## Quiz

What is the **sample rate** of an audio file that captures 96,000 samples per second?

- A. 44.1 kHz
- B. 48 kHz
- C. 96 kHz
- D. 192 kHz

## Quiz

Which of the following correctly describes **bit depth** in digital audio?

- A. The number of samples captured per second
- B. The number of amplitude levels captured per sample
- C. The number of bits used for file compression
- D. The file size in bits

## Quiz

If an audio file has a **bit depth** of **16** bits, how many possible **amplitude levels** can it represent?

- A. 16 levels
- B. 256 levels
- C. 65,536 levels
- D. 1,048,576 levels

## Quiz

An **audio file** has a sample rate of 44.1 kHz, a bit depth of 16 bits, and 2 channels (stereo). What is the **file size** for 1 minute of uncompressed audio?

- A. 176.4 KB
- B. 10.584 MB
- C. 5.292 MB
- D. 1.41 MB

## Quiz

An audio file with a **bit rate** of 128 kbps has a **duration** of 3 minutes. What is the **file size**?

- A. 2.88 MB
- B. 3.84 MB
- C. 4.32 MB
- D. 5.76 MB

## Quiz

An audio file has a **sample rate** of 96 kHz, a **bit depth** of 24 bits, **stereo** channels, and a **file size** of 50 MB. What is the **duration** of the audio?

- A. 3 minutes 5 seconds
- B. 4 minutes 10 seconds
- C. 5 minutes 30 seconds
- D. 6 minutes

## Quiz

A 5-minute audio file sampled at 44.100 samples per second. The amplitudes of the samples are quantized to 1.024 levels. Determine the size of the audio file?

- A. 132.3 Mb
- B. 13.23 Mb
- C. 2.205 Mb
- D. 220.500 bits

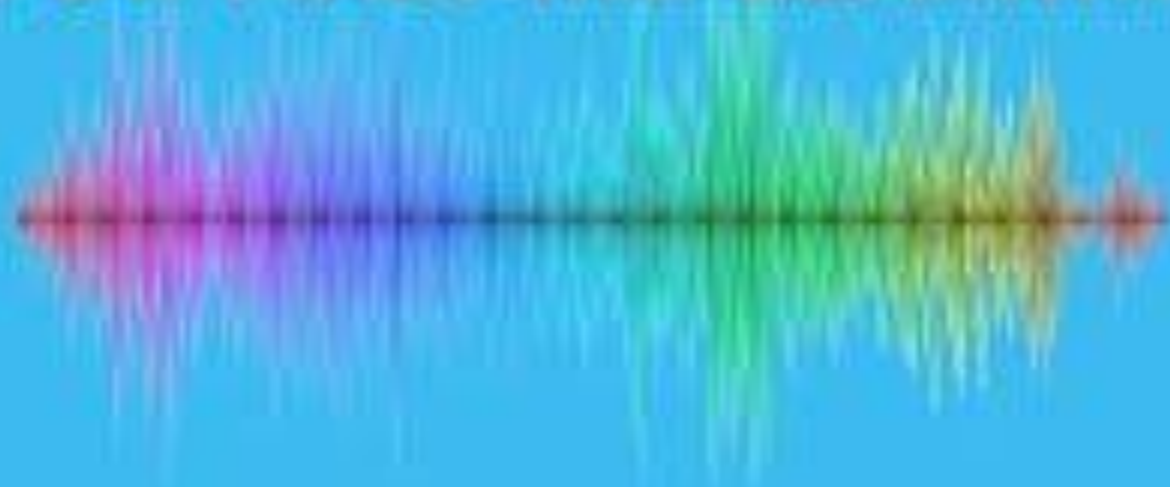
## 3.2. Signal Representation



## Waveform Basic Information

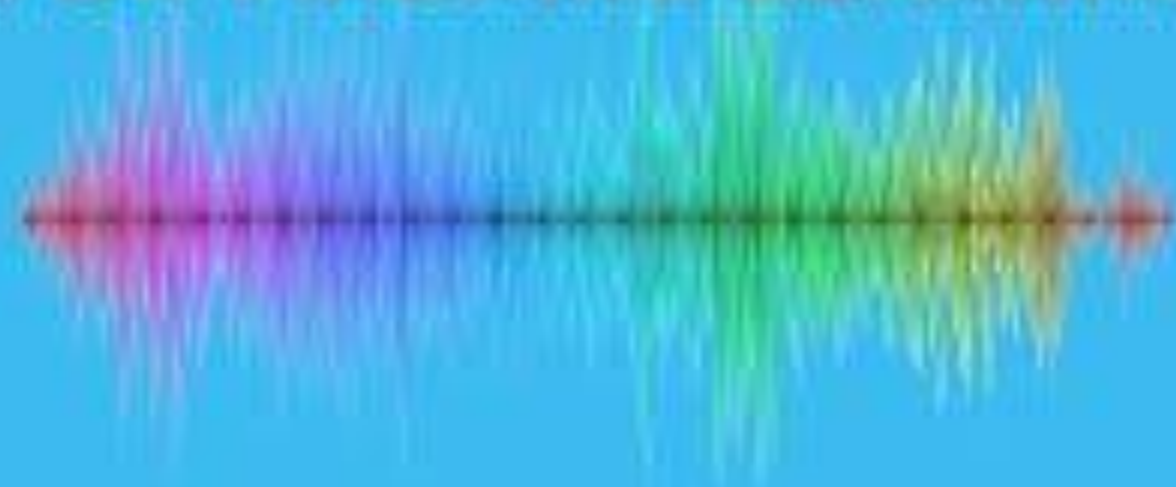
- Frequency
  - Refers to how "high" or "low" a sound is.
- Intensity
  - Refers to how "loud" or "soft" a sound is.
- Timbre
  - Refers to the unique "color" or "quality" of a sound that makes it distinguishable, even if two sounds have the same frequency and intensity.

# AUDIO SIGNAL PROCESSING FOR MACHINE LEARNING



## 2- SOUND AND WAVEFORMS

# AUDIO SIGNAL PROCESSING FOR MACHINE LEARNING

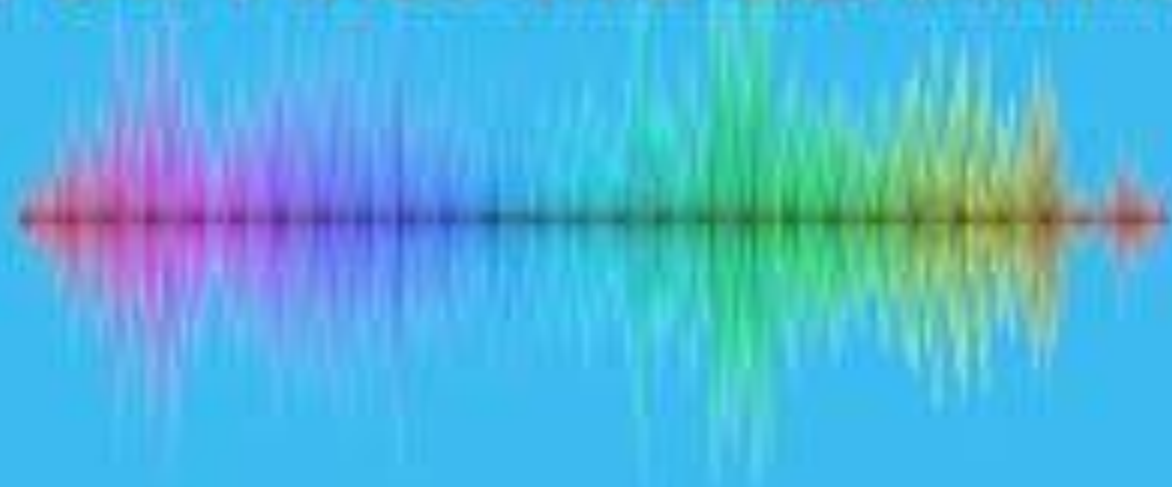


## 3- LOUDNESS AND TIMBRE

# Audio Signal Representation

- **Time** Domain Representation:
  - Represents a signal as it varies over time (e.g., an audio waveform)
- **Frequency** Domain Representation:
  - Describes a signal in terms of its frequency components using methods like the Fourier Transform.
- **Time-Frequency** Domain Representation:
  - Determines the signal's strength and timing characteristics

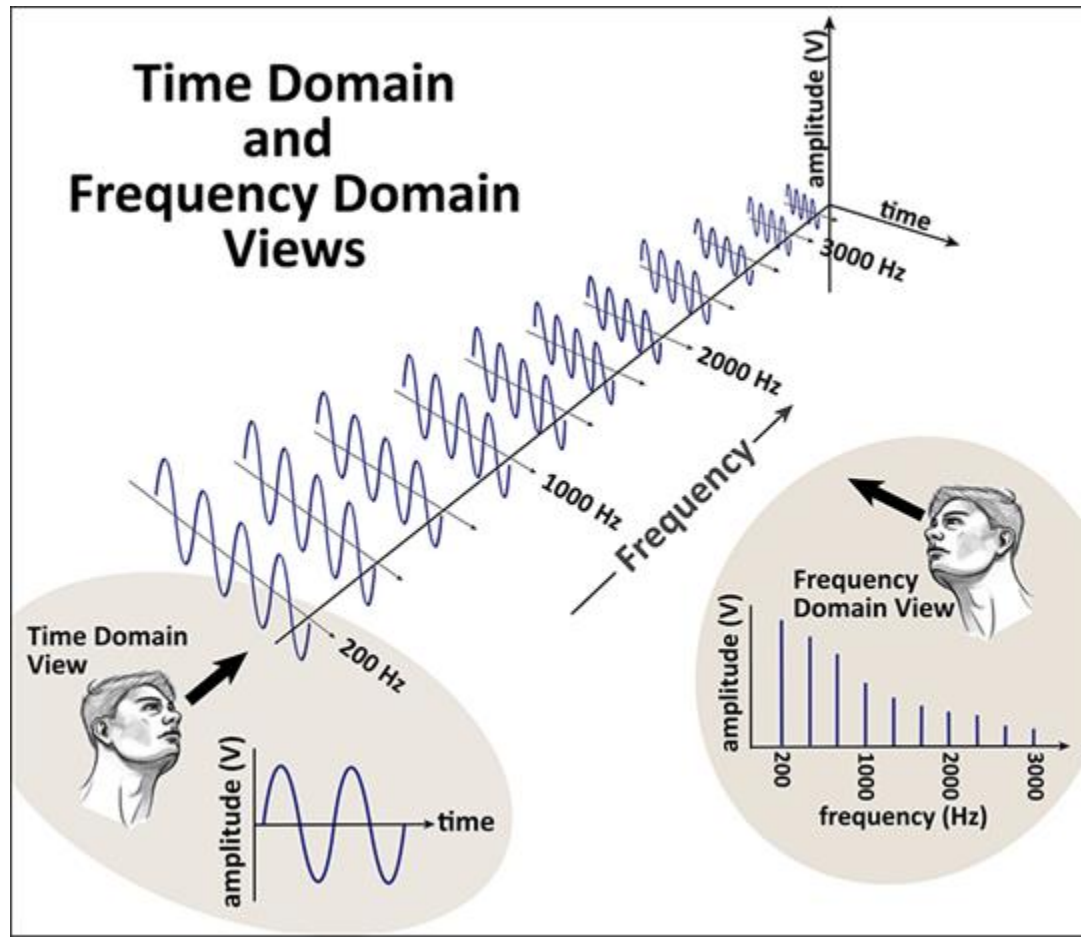
# AUDIO SIGNAL PROCESSING FOR MACHINE LEARNING



## 5- AUDIO FEATURES

# Time-Frequency Domain Representation

- Analyze and represent **signals** (e.g., audio) in both the **time** domain and the **frequency** domain **simultaneously**
  - How the **frequency content** of a **signal evolves**



# Time-Frequency Domain Methods

| Method                 | Resolution                  | Applications                  | Strengths                       | Weaknesses                          |
|------------------------|-----------------------------|-------------------------------|---------------------------------|-------------------------------------|
| <b>STFT</b>            | <b>Fixed time-frequency</b> | <b>Speech, music analysis</b> | <b>Simple, widely used</b>      | <b>Limited resolution trade-off</b> |
| Wavelet Transform      | Adaptive resolution         | Biomedical, transient signals | Good for transient signals      | Computationally complex             |
| Wigner-Ville Dist.     | High resolution             | Radar, quantum mechanics      | High time-freq resolution       | Cross-term interference             |
| <b>Mel Spectrogram</b> | <b>Mel scale resolution</b> | <b>Speech recognition, AI</b> | <b>Matches human perception</b> | <b>Limited raw frequency info</b>   |
| Constant-Q Transform   | Logarithmic resolution      | Music transcription           | Suited for pitch analysis       | Computationally intensive           |

# Short-Time Fourier Transform (STFT)

- Divides the signal into **small, overlapping time windows (segments)** and applies the **Fourier Transform** to each window
  - This creates a spectrogram
  - Provides a time-varying frequency spectrum
- **Pros:**
  - Easy to implement
  - Widely used and understood.
- **Cons:**
  - Fixed resolution: It uses the same window size for all frequencies, which limits the trade-off between time and frequency resolution.

Signal



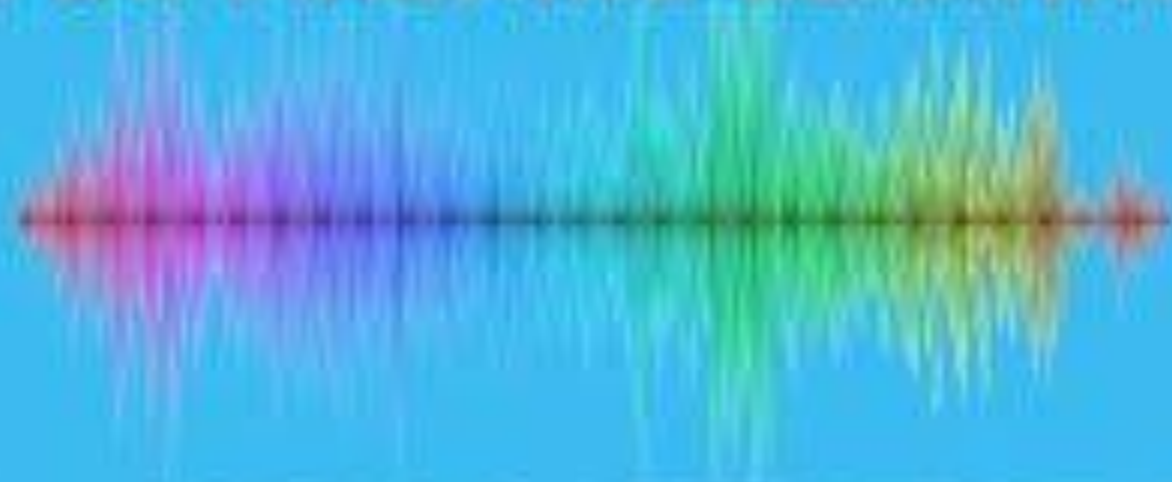
Winding



Transform



# AUDIO SIGNAL PROCESSING FOR MACHINE LEARNING

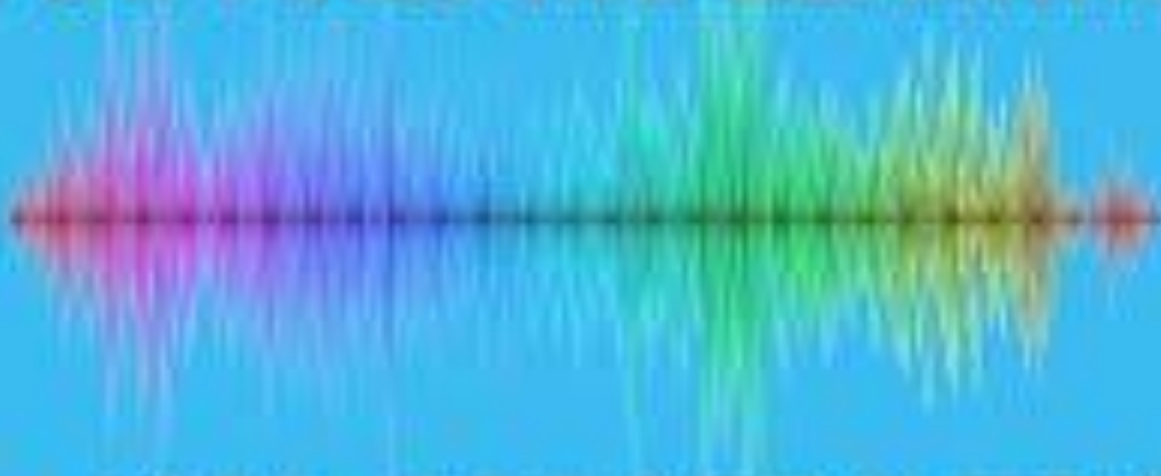


## 6- AUDIO FEATURES EXTRACTION PIPELINE

# Mel Spectrogram

- A **variant** of the **spectrogram** that maps frequencies onto the **Mel scale**, which **approximates** how **humans perceive pitch**.
  - A time-frequency representation where the y-axis is on the Mel scale
- Pros:
  - Matches human auditory perception.
  - Compact and efficient for machine learning tasks
- Cons:
  - Does not preserve raw frequency information.

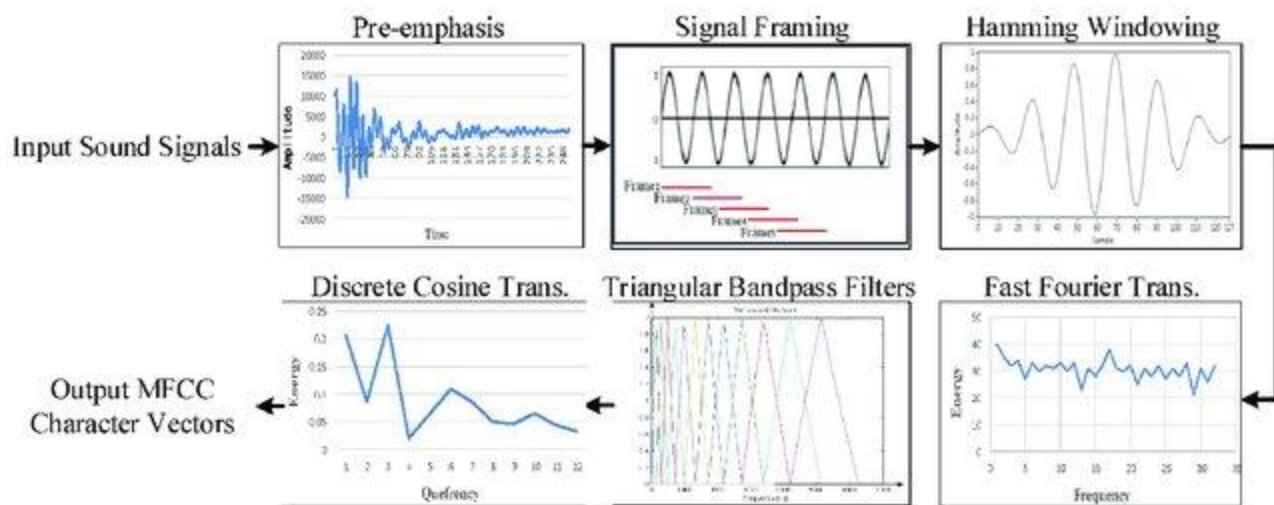
# AUDIO SIGNAL PROCESSING FOR MACHINE LEARNING



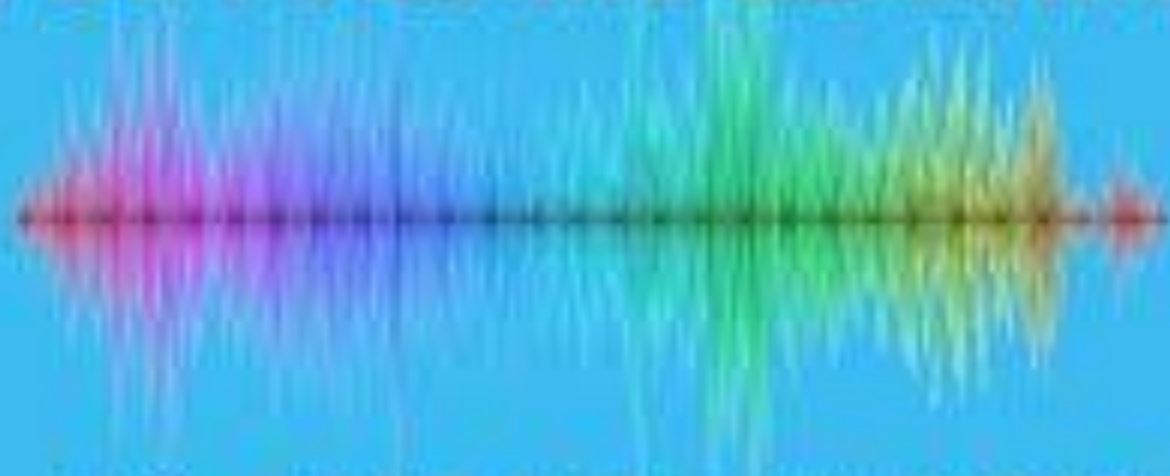
## 17 - MEL SPECTROGRAMS

# Mel-Frequency Cepstral Coefficients (MFCC)

- **Compact** representation of the **power spectrum** of an audio signal and are specifically designed to approximate how humans perceive sound (Mel scale)
  - Reduce high-dimensional audio data into **meaningful** and compact features aligned with **human perception**



# AUDIO SIGNAL PROCESSING FOR MACHINE LEARNING



## 19 - MFCCs: A KEY AUDIO FEATURE

## Homework

- Extract Mel Spectrogram using Python: [https://youtu.be/TdnVE5m3o\\_0?si=dio\\_XClWn-trMjWX](https://youtu.be/TdnVE5m3o_0?si=dio_XClWn-trMjWX)
- Extract MFCCs using Python: <https://youtu.be/WJI-17MNpdE?si=ptIJBZG6qPzldLWu>